Formalism: recovery limits, EFE, and tempered aggregation

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Formalism: recovery limits, EFE, and tempered aggregation

The primitives of sec. 8 are governed by a compact set of machine-checkable identities.

Formalism: recovery... (part 2)

Their counter is monotone across the methods and this section. Definitions 1 and 2 cover the posterior and cavity / PVI update (sec. 8.3).

Formalism: recovery... (part 3)

Lemma 3 gives the Rényi KL limit (sec. 8.6).

Formalism: recovery... (part 4)

Proposition 4 gives the β-loss and rcce NLL limits (sec. 8.7). Theorem 5 and Corollary 6 give belief-sharing and Bayes recovery (sec. 8.8).

Formalism: recovery... (part 5)

Proposition 7 below gives the expected-free-energy identity.

Each carries a tested residual. None grants a bounded-influence guarantee to the server-side `robust_aggregate` heuristic; that boundary is stated in sec. 6.3.

Recovery limits as the proof surface

The recovery limits separate client and server claims. The divergence and loss limits recover the standard-Bayes client update; the independently tested identity equates `robust_aggregate` with `log_linear_pool` when `robustness=0`, recovering the project's standard server pool.

Recovery limits as the... (part 2)

Under the explicit shared-support, posterior-log-potential, and fixed-weight assumptions in sec. 8.8, that pool is a categorical specialization of Eq. 7's message-combination term, not recovery of the complete source protocol (Friston et al. 2024).

Recovery limits as the… (part 3)

We collect the five residuals that pin those limited claims, each emitted by the test-suite and reported in sec. 10.1, never hardcoded.

Recovery limits as the... (part 4)

Read each row of the table below as a triple: the robust primitive, the trusting knob value at which it must collapse onto its standard-Bayes or project-local counterpart, and the tested residual measuring whatever gap survives at that value. The rows differ in what kind of check they are.

Recovery limits as the... (part 5)

The divergence and loss rows evaluate inside the implementation's closed-form switch band, so their zeros are exact branch identities — guaranteed by construction, not measurements that could have come out otherwise.

Recovery limits as the... (part 6)

Their genuine falsifiers are the *off-switch* convergence residuals, evaluated just outside the band (at $\alpha = 1.00001$, $\beta = 1.00 \times 10^{-6}$, $q_{\text{loss}} = 1.00 \times 10^{-6}$) where the general formulas run and a nonzero gap is possible: those residuals are 1.66×10^{-5}, 1.24×10^{-5},

Recovery limits as the... (part 7)

and 1.12×10^{-5} respectively, and any failure of those quantities to shrink toward the limit would falsify the containment claim.

Recovery limits as the... (part 8)

The posterior row is a measured identity on the general code path, so its near-zero residual is itself the falsification surface. The aggregate row's zero at exactly $c = 0$ is likewise branch-exact; the identity is additionally exercised on the iterative code path at near-zero robustness,

Recovery limits as the… (part 9)

where the consensus must still land on the log-linear pool to tight tolerance.

Recovery limits as the… (part 10)

Identity (owner statement)	Trusting limit	Tested residual
Rényi $\to$ KL (Lemma 3, eq. 6)	$\alpha \to 1$	0

Recovery limits as the... (part 11)

The central project identity is the aggregation collapse of eq. 10: `robust_aggregate(robustness=0)` equals the log-linear pool eq. 9. Its source bridge is deliberately narrow.

Recovery limits as the... (part 12)

Take the admitted posteriors defined in sec. 8.8, on a finite common support with $q_n(s) > 0$, and represent each Eq. 7 softmax input as a posterior log potential $m_n(s) = \log q_n(s) + \kappa_n$, with κ_n constant in s and fixed declared weights w_n independent of the emerging consensus.

Recovery limits as the... (part 13)

Softmax then cancels the additive constants and yields the project pool.

Recovery limits as the... (part 14)

This identifies only the source equation's message-combination term; it does not identify source message construction, cavity/exclusion policy, scheduling, generative factors, or the complete protocol.

Recovery limits as the... (part 15)

Theorem 5 (sec. 8.8) states that specialization and the local $c = 0$ identity; the residual 0 above pins the latter.

Recovery limits as the... (part 16)

Corollary 6 establishes the separate client result: `generalized_posterior(KLD, NLL)` reproduces the closed-form prior-times-likelihood Bayes posterior of eq. 11 to residual 5.55e-17.

Recovery limits as the... (part 17)

Pooling such local posteriors has the stated categorical specialization only under the theorem's assumptions.

Recovery limits as the... (part 18)

The honesty contract binds here: the theorem and corollary cover only the recovery identity and the per-agent rigorous axis (Proposition 4); no statement transfers the bounded-influence guarantee to the server-side divergence-reweighting heuristic,

Recovery limits as the... (part 19)

whose positive property is the $\texttt{robustness} = 0$ limit of eq. 10. A scoped no-go rejects a declared separable objective class without certifying another.

Expected-free-energy identity as an algebraic check

The active-inference substrate that drives the studies of sec. 10 is a categorical specialization of the expected-free-energy algebra discussed by Friston et al. (Friston et al. 2024).

Expected-free-energy... (part 2)

It decomposes the expected free energy of a policy π into two equivalent two-term forms.

Expected-free-energy... (part 3)

The risk-plus-ambiguity (cost) view and the negated pragmatic-plus-epistemic (value) view are the same scalar $G(\pi)$ rearranged (Da Costa et al. 2020; Friston et al. 2024):

Expected-free-energy... (part 4)

$$
\begin{aligned}
G(\pi) &= \underbrace{\text{risk} + \text{ambiguity}}_{\text{cost view}} \\
&= -\underbrace{(\text{pragmatic} + \text{epistemic})}_{\text{value view}} .
\end{aligned}
\tag{22}
$$

Expected-free-energy… (part 5)

The two views are not approximations of one another; they are the same scalar rearranged. In the implementation the shared entropy term enters both sides of the rearrangement, so the identity residual is zero by construction — it is a definitional consistency check on the decomposition's bookkeeping,

Expected-free-energy... (part 6)

not an independent measurement.

The scientific content lives in the per-term semantics, which are pinned independently of the identity (see the closing clause of the proposition below):

Expected-free-energy... (part 7)

$$\begin{aligned}&(\text{risk} + \text{ambiguity}) \\ &\quad + (\text{pragmatic} + \text{epistemic}) \equiv 0.\end{aligned} \tag{23}$$

Expected-free-energy… (part 8)

Proposition (Expected-free-energy decomposition identity)

For the categorical model in `expected_free_energy.py`*, the cost and negated-value forms of (22) give the same* $G(\pi)$*; hence (23) holds to machine precision. Cross-entropy and entropy splits establish the equality.*

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Expected-free-energy... (part 9)

The residual is pinned to 10^{-9}. Independent checks require deterministic likelihoods to give zero ambiguity, uninformative likelihoods zero epistemic value, and preference-matched predictions lower risk.

Expected-free-energy... (part 10)

Here risk is $\mathrm{KL}(q(o \mid \pi) \,\|\, p_C(o))$, and ambiguity is expected likelihood entropy $\mathbb{E}_{q(s)}[H[p(o \mid s)]]$.

Expected-free-energy... (part 11)

Pragmatic value is expected log-preference $\mathbb{E}_{q_\pi(o)}[\ln p_C(o)]$. Epistemic value is the state-outcome mutual information $H[q_\pi(o)] - \mathbb{E}_{q(s)}[H[p(o \mid s)]]$.

Expected-free-energy... (part 12)

The executed formal-specialization diagnostic of fig. 11 uses a uniform prior over the nine locations. This is intentional: the canonical sentinel-world D_0 is a point mass at the den,

Expected-free-energy... (part 13)

and under that fully resolved prior the mutual-information term is zero because there is no state uncertainty for an observation to reduce.

Expected-free-energy... (part 14)

The uncertainty-bearing diagnostic makes the epistemic term visible without changing the canonical D_0 used by the inference and recovery studies. Thus a near-zero epistemic value is a meaningful null condition, not a missing term.

Expected-free-energy... (part 15)

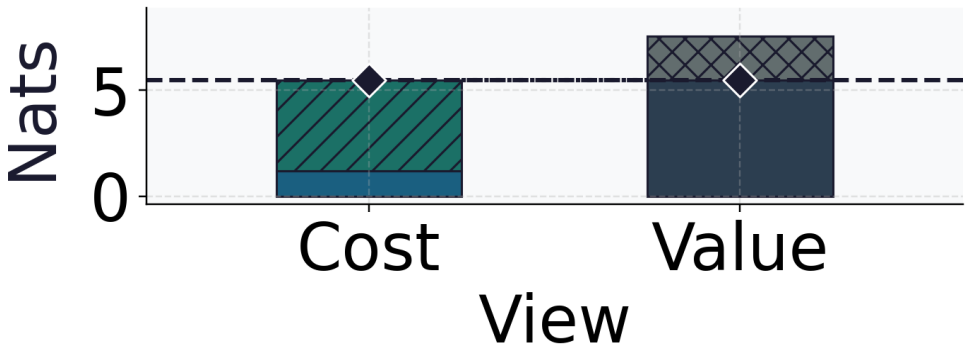
fig. 11 shows the additive risk-plus-ambiguity view beside a signed pragmatic/epistemic waterfall whose terminal endpoint is $G(\pi)$, labels epistemic value as $I(s; o \mid \pi)$, and annotates the identity residual; it visualizes Proposition 7,

Expected-free-energy... (part 16)

not a fitted result, so it carries no error bars.

Expected-free-energy… (part 17)

Cost and signed value views



Expected-free-energy… (part 18)

Cost: risk 1.187 plus ambiguity 4.271 nats. Hatched upper segment is ambiguity.

Expected-free-energy… (part 19)

Value: −pragmatic 7.534 plus −epistemic -2.077 nats. Crosshatching marks the signed epistemic correction.

Expected-free-energy… (part 20)

Terminal diamond: G = 5.457 nats. The right endpoint equals G; the intermediate top need not equal G.

Expected-free-energy… (part 21)

Identity residual: 0.000e+00 nats. G = risk + ambiguity = −(pragmatic + epistemic).

Expected-free-energy... (part 22)

Uniform diagnostic prior makes information gain visible. One categorical identity diagnostic; no stochastic replication or uncertainty interval.

Expected-free-energy... (part 23)

The expected-free-energy identity of eq. 23 is the action-selection counterpart of the inference-side recovery limits collected above: both are exact, closed-form, machine-checkable identities over the same categorical generative model.

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Expected-free-energy... (part 24)

Together they establish that the FedGVI-federated active-inference colony of this work is built on verified algebra throughout — the per-agent generalized-Bayes update recovers standard Bayes, the aggregation recovers the project log-linear pool under its qualified categorical bridge,

Expected-free-energy... (part 25)

and the policy scoring decomposes exactly — so every robustness result in sec. 10 is a controlled departure from a known, tested fixed point rather than an unmoored claim.

Tempered aggregation free energy and the accuracy-guarantee trade

The recovery limits and the expected-free-energy identity fix the *endpoints* of the aggregation family; the remaining formal question is what a controlled departure from the unit-entropy server buys and what it costs.

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Tempered aggregation free... (part 2)

The objective-backed variational aggregator of sec. 15 holds its consensus-entropy term at unit weight.

Tempered aggregation free... (part 3)

Freeing that single coefficient produces a one-parameter family whose only moving part is the sharpness of the consensus, and whose raw effective-weight bound is provably untouched. The following proposition isolates exactly that separation — algebra that moves versus algebra that does not —

Tempered aggregation free... (part 4)

before the interpretation subsections turn to the empirical accuracy question the algebra cannot settle on its own.

Tempered aggregation free... (part 5)

Proposition (Tempered aggregation free energy)

Let $\lambda > 0$ and let F_λ be (30). At fixed a, the q-minimizer is (31). At $\lambda = 1.0$, F_λ and both updates equal the standard variational aggregate bit-for-bit. The a-update and bound $a_n \leq w_n$ omit λ and hold for every $\lambda > 0$.

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Tempered aggregation free… (part 6)

At that default, the endpoint-selection rule also reduces bit-for-bit to the standard variational aggregate (Definition 10; sec. 15).

Tempered aggregation free... (part 7)

The proposition is intentionally more specific than the phrase “temperature improves robustness.” It identifies exactly which part of the variational server changes when the entropy coefficient changes, and it separates that algebra from the empirical accuracy question.

Tempered aggregation free... (part 8)

The objective and its update rules are a generalized-Bayes construction in the sense of (Bissiri et al. 2016; Knoblauch et al. 2022), while the particular client/server decomposition is the one implemented and tested here.

What the entropy weight controls

For a fixed effective-weight vector a and $\lambda > 0$, the q-block in eq. 31 is a weighted geometric pool of the local posteriors with inverse temperature $1/\lambda$. Lower λ concentrates more sharply on states that receive consistent log-belief support;

What the entropy weight... (part 2)

larger λ spreads probability mass and retains more entropy.

What the entropy weight... (part 3)

As $\lambda \downarrow 0$, the positive-temperature expression approaches a winner-take-most consensus (subject to ties and finite numerical support), whereas large λ approaches a flatter distribution.

What the entropy weight... (part 4)

The implementation exposes that endpoint as a separately defined deterministic tied-argmax rule; it does not substitute $\lambda = 0$ into the objective or coordinate update.

This is a controlled change in the consensus geometry, not an automatic outlier detector.

What the entropy weight... (part 5)

The coupling matters. Although the formula for the a-block does not contain λ, the fixed point can still change because a_n is evaluated at the new q.

What the entropy weight... (part 6)

The correct statement is therefore conditional: for any current consensus, the effective-weight update and the raw bound $a_n \leq w_n$ are unchanged; after alternating updates, different temperatures can reach different coupled (q, a) fixed points.

What the entropy weight... (part 7)

This distinction prevents the temperature result from being read as a theorem that the normalized influence or accuracy is invariant in λ.

Recovery at the qualified log-linear-pool corner

At the configured default $\lambda = 1.0$, the entropy coefficient is the unit coefficient used by the original variational aggregator. The implementation therefore recovers that aggregator bit-for-bit,

Recovery at the qualified… (part 2)

including its block updates and its endpoint-selection rule.

Recovery at the qualified… (part 3)

Turning the robustness strength c to zero then sets every server weight to its base value and gives the tempered log-linear pool. At the default temperature this is the ordinary log-linear pool of eq. 9.

Recovery at the qualified… (part 4)

Under the shared-support, posterior-log-potential, and fixed-weight assumptions of sec. 8.8, that is the categorical specialization of Eq. 7's message-combination term; it is not the complete belief-sharing protocol of (Friston et al. 2024). Away from that default,

Recovery at the qualified… (part 5)

the result is a tempered generalization of the pool and should not be described as Friston's Eq. 7 itself.

Recovery at the qualified... (part 6)

This nested limit is useful for interpretation. The $c \to 0$ limit identifies the aggregation family with a known consensus operator; the $\lambda = 1.0$ slice identifies the objective-backed implementation used in the main server comparison.

Recovery at the qualified… (part 7)

Neither limit grants the server-side `robust_aggregate` heuristic a variational objective or a bounded-influence guarantee. The FedGVI literature's rate and robustness results remain attached to their stated loss, divergence,

Recovery at the qualified… (part 8)

and sampling assumptions (Mildner, Hamelijnck, et al. 2025; Mildner, Giampouras, et al. 2025).

What the accuracy–guarantee trade can establish

The executed grid over $\lambda \in \{0.1, 0.2, 0.3, 0.5, 0.7, 1\}$ is a finite sensitivity study, not a search over a continuous optimum.

What the... (part 2)

It asks whether any tested temperature narrows the point-accuracy gap to the sharp heuristic while retaining the same effective-weight update. The closest tested temperature is $\lambda^* = 0.3$,

with observed gap 0.0008.

What the... (part 3)

These tokens are computed from the executed contaminated-colony trials and are reported with the grid definition so a reader can reproduce the selection rule.

What the... (part 4)

The result has two distinct readings. If a lower temperature improves the paired point-accuracy comparison, it provides design evidence that entropy regularization can be tuned rather than accepted as a fixed conservatism penalty.

What the... (part 5)

If no tested temperature closes the gap, that negative result is still informative: within this objective family, the same entropy mechanism that keeps consensus diffuse can limit exact point recovery under confident contamination.

What the… (part 6)

In either case, the grid does not identify a universally best temperature, establish minimax robustness, or transfer the variational raw weight bound to a different estimator. Generalized-Bayes calibration and robust-loss theory motivate the family,

What the... (part 7)

but only the executed categorical design supports the present finite-grid statement (Bissiri et al. 2016; Knoblauch et al. 2022; Mildner, Hamelijnck, et al. 2025).

Publication-facing interpretation

The practical contract is consequently three-part. Use the default temperature when exact compatibility with the tested variational server is the priority.

Publication-facing... (part 2)

Explore the declared grid when the application can trade concentration against the same raw effective-weight control.

Publication-facing... (part 3)

Treat any selected temperature as a configuration-specific empirical choice until it is tested under a new contamination mechanism, colony size, sensor model, or loss. This is the appropriate bridge between the formal objective and the active-inference setting:

Publication-facing... (part 4)

it exposes a tunable consensus geometry while keeping recovery, guarantee, and accuracy claims on separate evidence tracks.